

CURVE TABLE

CURVE	RADIUS	ARC LTH	CHD LTH	CHD BEARING	DELTA
C1	5364.91	300.59	300.55	S86°27'07"E	1°13'37"
C1	5379.81	21.41	21.41	S84°57'30"E	0°13'41"
C1	5379.91	112.28	112.27	S85°40'22"E	1°11'45"
C1	5364.91	132.42	132.42	S85°33'15"E	1°24'51"
C1	5364.91	30.05	30.05	S86°25'18"E	0°19'15"
C1	1220.09	30.04	30.04	S86°44'33"E	0°14'18"
C1	5364.91	108.08	108.08	S87°28'48"E	1°09'15"
C1	280.00	37.20	37.17	S85°51'28"E	7°36'43"
C2	280.00	49.41	49.33	S76°59'47"E	1°06'40"
C3	250.00	77.33	77.02	S80°48'08"E	1°43'23"
C4	220.00	30.55	30.52	S85°41'10"E	7°57'19"
C4	220.00	37.51	37.46	S76°49'28"E	1°04'04"
C5	280.00	27.00	27.01	N15°17'42"E	5°31'44"
C7	280.00	59.59	59.48	S06°26'W	12°11'39"
C8	250.00	77.33	77.02	N09°11'52"E	1°43'23"
C9	220.00	34.69	34.63	N13°32'31"E	9°02'04"
C10	220.00	33.36	33.33	N04°40'50"E	8°41'19"
C11	280.00	19.05	19.04	S16°06'37"W	3°53'52"
C12	280.00	63.97	63.83	S07°37'W	1°05'22"
C13	280.00	3.93	3.93	S00°42'15"W	0°44'08"
C14	250.00	77.33	77.02	S09°11'52"W	1°43'23"
C15	220.00	27.05	27.03	S14°32'15"W	7°02'37"
C16	220.00	41.01	40.93	S05°40'34"W	1°04'46"
C17	1220.00	50.23	50.22	S73°07'13"E	2°21'32"
C18	1220.00	73.90	73.89	S76°02'06"E	3°28'14"
C19	1220.00	16.27	16.27	S78°09'08"E	0°45'51"
C20	1250.00	143.83	143.77	S05°38'25"E	1°03'28"
C21	1250.00	29.54	29.54	S72°37'04"E	1°21'14"
C22	1250.00	114.31	114.27	S75°54'52"E	5°14'22"
C23	280.00	26.17	26.16	S15°03'52"W	5°21'17"
C24	280.00	58.89	58.76	S02°33'24"E	1°02'03"
C25	250.00	75.95	75.65	S09°02'20"W	1°24'20"
C26	220.00	40.73	40.67	N05°38'25"E	10°36'28"
C27	220.00	28.10	28.09	N14°20'34"E	6°17'52"
C28	230.00	44.68	44.61	S84°05'57"E	1°10'46"
C29	200.00	36.85	36.79	S84°05'57"E	1°10'46"
C30	170.00	33.02	32.97	S84°05'57"E	1°10'46"
C31	180.00	36.80	36.74	S72°40'37"E	1°14'04"
C32	180.00	13.69	13.68	S64°38'27"E	4°21'25"
C33	180.00	63.51	63.18	S52°21'14"E	20°13'01"
C34	150.00	95.00	93.42	S02°33'24"E	36°17'20"
C35	120.00	76.00	74.74	S60°23'24"E	36°17'20"
C36	130.00	37.16	37.03	S70°20'47"E	16°22'32"
C37	10.00	12.23	11.48	N82°47'53"E	7°05'13"
C38	100.00	55.65	54.93	S62°39'35"E	31°52'56"
C39	10.00	22.04	17.84	S15°23'24"E	126°17'20"
C40	120.00	3.84	3.84	S00°34'53"E	1°50'07"
C41	120.00	85.34	83.53	S12°52'20"E	40°44'47"
C42	150.00	111.48	108.93	S02°57'17"E	42°34'54"
C43	180.00	38.99	38.91	N36°02'24"W	12°24'39"
C44	180.00	66.43	66.06	N19°15'41"W	21°08'47"
C45	180.00	27.35	27.32	N04°08'43"W	8°53'48"
C46	330.00	90.51	90.23	S12°08'E	15°42'53"
C47	330.00	34.22	34.21	S22°55'42"E	5°56'50"
C48	300.00	137.37	136.17	S12°46'53"E	28°14'07"
C49	270.00	91.24	90.72	S15°34'52"E	20°38'09"
C50	220.00	41.55	41.49	S20°29'17"E	10°49'20"
C51	220.00	60.77	60.58	S07°09'48"E	15°49'36"
C52	250.00	112.28	112.23	S12°34'28"E	24°38'56"
C53	280.00	130.23	129.06	S12°34'28"E	26°38'57"
C54	270.00	11.91	11.91	N89°04'21"E	2°31'40"
C55	270.00	108.20	107.48	N76°19'40"E	22°57'42"
C56	270.00	80.55	80.25	N61°19'03"E	17°05'37"
C57	300.00	222.96	217.86	N69°02'43"E	42°34'54"
C58	330.00	245.28	239.65	S69°02'43"W	42°34'54"

LINE TABLE

#	BEARING	DISTANCE
L1	N11°27'57"E	113.79'
L2	N11°27'57"E	110.92'
L3	N11°27'57"E	107.93'
L4	N11°27'57"E	107.44'
L5	N11°27'57"E	106.98'
L6	N11°27'57"E	101.69'
L7	N47°45'16"E	7.97'
L8	S42°14'44"W	105.50'
L9	N42°14'44"W	105.50'
L10	N42°14'44"W	105.50'
L11	N42°14'44"W	105.50'
L12	S47°45'16"W	30.00'
L13	S47°49'09"W	23.64'
L14	N42°14'44"W	105.50'
L15	N42°14'44"W	105.50'
L16	N42°14'44"W	105.50'
L17	N42°14'44"W	105.50'
L18	S89°39'50"W	37.50'
L19	N42°14'44"W	105.50'
L20	N42°14'44"W	105.50'
L21	N42°14'44"W	105.50'
L22	N11°27'57"E	35.84'
L23	S11°27'57"W	105.50'
L24	N78°32'03"W	99.22'
L25	S00°20'10"W	102.89'
L26	N89°39'50"W	105.50'
L27	N89°39'50"W	105.50'
L28	N11°27'57"E	105.50'
L29	N11°27'57"E	105.50'
L30	N11°27'57"E	105.50'
L31	N11°27'57"E	105.50'
L32	N11°27'57"E	105.50'
L33	S78°32'03"E	22.03'
L34	N11°27'57"E	105.50'
L35	N42°14'44"W	105.50'
L36	N47°45'16"E	72.43'
L37	S47°45'16"W	229.00'
L38	N42°14'44"W	105.50'
L39	N27°32'13"E	92.47'
L40	S00°20'10"W	76.12'
L41	S00°20'10"W	76.12'
L42	S00°20'10"W	76.12'
L43	S00°20'10"W	76.12'
L44	S00°20'10"W	76.12'
L45	S42°14'44"E	65.53'
L46	N81°26'37"E	61.61'
L47	S00°20'10"W	70.00'
L48	S00°20'10"W	70.00'
L49	S00°20'10"W	70.00'
L50	S00°20'10"W	70.00'
L51	S00°20'10"W	70.00'
L52	S00°20'10"W	70.00'
L53	S00°20'10"W	70.00'
L54	S00°20'10"W	70.00'
L55	S00°20'10"W	70.00'
L56	S00°20'10"W	75.71'
L57	S89°39'50"E	45.21'
L58	S00°20'10"W	70.00'
L59	S00°20'10"W	70.00'
L60	S00°20'10"W	70.00'
L61	S00°20'10"W	70.00'
L62	S00°20'10"W	70.00'
L63	S00°20'10"W	70.00'
L64	S00°20'10"W	70.00'
L65	S00°20'10"W	70.00'
L66	S00°20'10"W	70.00'
L67	S00°20'10"W	70.00'
L68	S00°20'10"W	70.00'
L69	S00°20'10"W	75.71'
L70	S00°20'10"W	3.66'



BOUNDARY DESCRIPTION

PART OF THE SOUTHEAST QUARTER OF SECTION 5, THE SOUTHWEST QUARTER OF SECTION 5, THE NORTHEAST QUARTER OF SECTION 7 AND THE NORTHWEST QUARTER OF SECTION 8, TOWNSHIP 17 SOUTH, RANGE 6 WEST, SALT LAKE BASE AND MERIDIAN, U.S. SURVEY, MORE PARTICULARLY DESCRIBED AS FOLLOWS:

BEGINNING AT A POINT, SAID POINT BEING 2217.24 FEET SOUTH 89°04'37" WEST AND 40.46 FEET NORTH 00°55'23" WEST FROM THE SOUTH QUARTER CORNER OF SAID SECTION 5; THENCE SOUTH 47°31'21" WEST 61.00 FEET; THENCE SOUTH 89°04'37" WEST 316.53 FEET; THENCE SOUTH 47°37'02" EAST 210.11 FEET; THENCE SOUTH 47°42'19" WEST 191.84 FEET; THENCE NORTH 89°15'00" WEST 1468.30 FEET; THENCE NORTH 00°09'19" EAST 243.94 FEET; THENCE NORTH 00°22'27" EAST 130.14 FEET; THENCE SOUTH 84°28'55" EAST 392.39 FEET; THENCE SOUTH 72°23'40" EAST 788.98 FEET; THENCE SOUTH 77°11'54" EAST 281.56 FEET; THENCE SOUTH 78°19'35" EAST 661.92 FEET; THENCE SOUTH 47°45'16" WEST 711.69 FEET; THENCE SOUTH 42°11'35" EAST 400.00 FEET TO THE POINT OF BEGINNING.

CONTAINING 2350036 SQUARE FEET OR 53.949 ACRES.

LEGEND

- SECTION CORNER
- BOUNDARY LINE
- LOT LINE
- ADJOINING PROPERTY
- CENTERLINE
- 10' PUBLIC UTILITY EASEMENT
- SECTION TIE LINE



VICINITY MAP

NO SCALE



REVISIONS	DESCRIPTION
DATE	

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Delta Crossings
PART OF THE SECTIONS 5, 6, & 8, T.17S., R.6W., S.L.B. & M., U.S. SURVEY
DELTA CITY, WILLARD COUNTY, UTAH

Preliminary Plan

Developer:

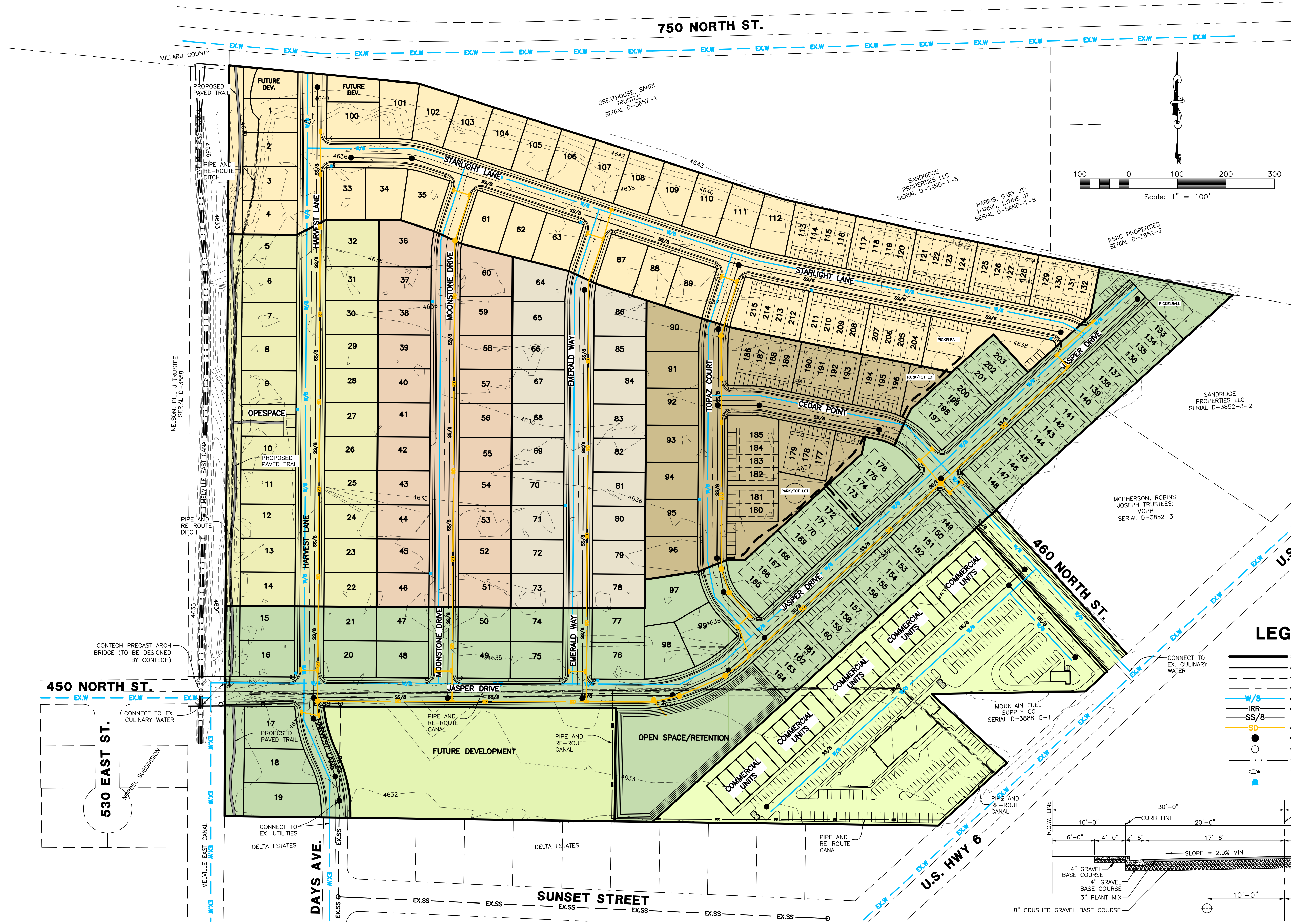
Bill Souvall
Souvall Brothers
3701 S. 700 W.
Salt Lake City, Utah, 84119
PH:(801) 808-1667

SITE DATA

TOTAL PROJECT AREA:	53.95 ACRES
TOTAL RESIDENTIAL AREA:	44.74 ACRES
TOTAL RESIDENTIAL LOTS:	109 LOTS
TOTAL TOWNHOME UNITS:	102 UNITS
DENSITY:	4.7 UNITS PER ACRES
REQUIRED OPENSACE FOR TOWNHOME LOTS	28 STALLS
102 UNITS @ 500 S.F./UNIT = 51,000 S.F. (1.17 ACRES)	
TOWNHOME OPENSACE PROPOSED:	1.90 ACRES
TOTAL TOTAL COMMERCIAL AREA:	5.78 ACRES
TOTAL RESIDENTIAL UNITS PARKING:	90 STALLS
TOTAL RESIDENTIAL ON-STREET PARKING:	280 STALLS
TOTAL COMMERCIAL PARKING:	398 STALLS (TOTAL)

Delta Crossings

Delta Crossings, Millard County, Utah



Storm Runoff Calculations

The following runoff calculations are based on the Rainfall - Intensity - Duration Frequency Curve for the Delta, Utah taken from the NOAA Atlas 14 database. Calculations have been completed for the 10yr 24-hr storm event. Storm water runoff has been calculated for a fully developed site and full retention.

The calculations are as follows:

Drainage Area:	54.19 acres or 2,360,516 ft ²
Runoff Coefficients:	
Paved Area	429,842
Single Family	214,600
Townhome	194,870
Commercial	186,152
Landscape Area	1,460,253
Weighted Runoff Coefficient:	C = 0.9
	C = 0.9
	C = 1.9
	C = 2.9
	C = 0.2
	C = 0.80

LID Retention:	80" Percentile Rainfall Event (I)	Yes	0.45 in
	Is the site location for LID?		
	Site Imperviousness (I)	0.27	
	NRCS Soil Group	C-19	
	R ₁ Equation	0.834*1.22	
	R ₂ (Soil Group A: 0.84*1.302; B: 0.84*1.189; C/D: 0.83*1.122)	0.19	
	V ₁₀₀ = R ₁ x R ₂ x Total Site SF	17180	c.f.

Rainfall Intensity:	10-yr intensity for a 30 minute storm - Pipe Capacity	1.06	in/hr
Peak Runoff:			
Runoff Coefficient	C =	0.80	
Rainfall Intensity	I =	1.06 in/hr	
Area	A =	54.19 ACRES	
Q	Q =	34.44 cfs	

Percolation Rate:			
Drainage Area	15	s.f.	
Percolation	15	in/hr	
Percolation out	(15in/hr) * (1512in/1hr) * (1min/60sec) * (60min/1hr) * (1hr/1hr)	1512	in

Volume of Runoff for 10-year Storm Event:			
C =	0.80		
I =	See Below in/hr		
A =	2360516.00 ft ²		
Q =	0.00 cfs		

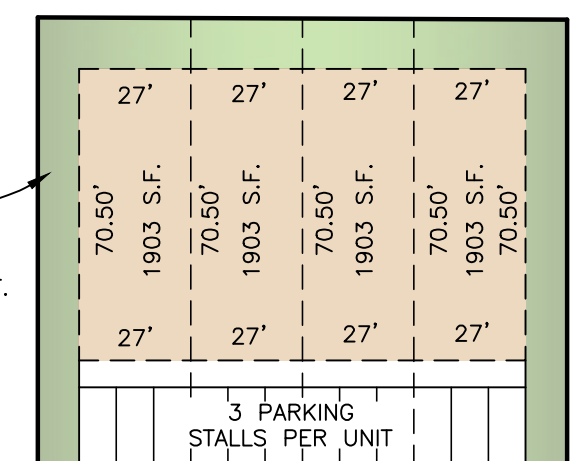
Time (min)	Q (cfs)	Vol. in (ft ³)	Vol. out (ft ³)	Difference (ft ³)
0	0.00	0.00	0	0
5	3.00	81.00	24071	0
10	6.00	162.00	37347	0
15	9.00	243.00	46291	0
20	12.00	324.00	52267	0
25	15.00	405.00	56660	0
30	18.00	486.00	59520	0
35	21.00	567.00	60977	0
40	24.00	648.00	61020	0
45	27.00	729.00	60001	0
50	30.00	810.00	57955	0
55	33.00	891.00	54901	0
60	36.00	972.00	50930	0
65	39.00	1053.00	46143	0
70	42.00	1134.00	40651	0
75	45.00	1215.00	34560	0
80	48.00	1296.00	27971	0
85	51.00	1377.00	20980	0
90	54.00	1458.00	13690	0
95	57.00	1539.00	6000	0
100	60.00	1620.00	0	0

Orifice Sizing:			
Given:	Q =	0.00	cfs
	H =	64.4	ft
	R =	3.00	ft
	Cd =	0.62	
	R =	SCORTCHG (70.50' x 70.50')	
	D =	0.00	ft
	A =	0.00	inches ²
		0.0000	ft ²

Aboveground Basin Sizing:			
Length	350		
Width	135		
Depth	3		
Top Area	47250		
Bottom Area	38844		
Average Area	43047		
Volume	129141		

Pipe Sizing:			
Description	Pipe Size	Slope	Cap. (cfs)
30" RCP	36	0.30%	38.88
			34.44

SUMMARY: The required 100-yr storage volume is 161,340 cubic feet. Orifice size is 0.0 inches.



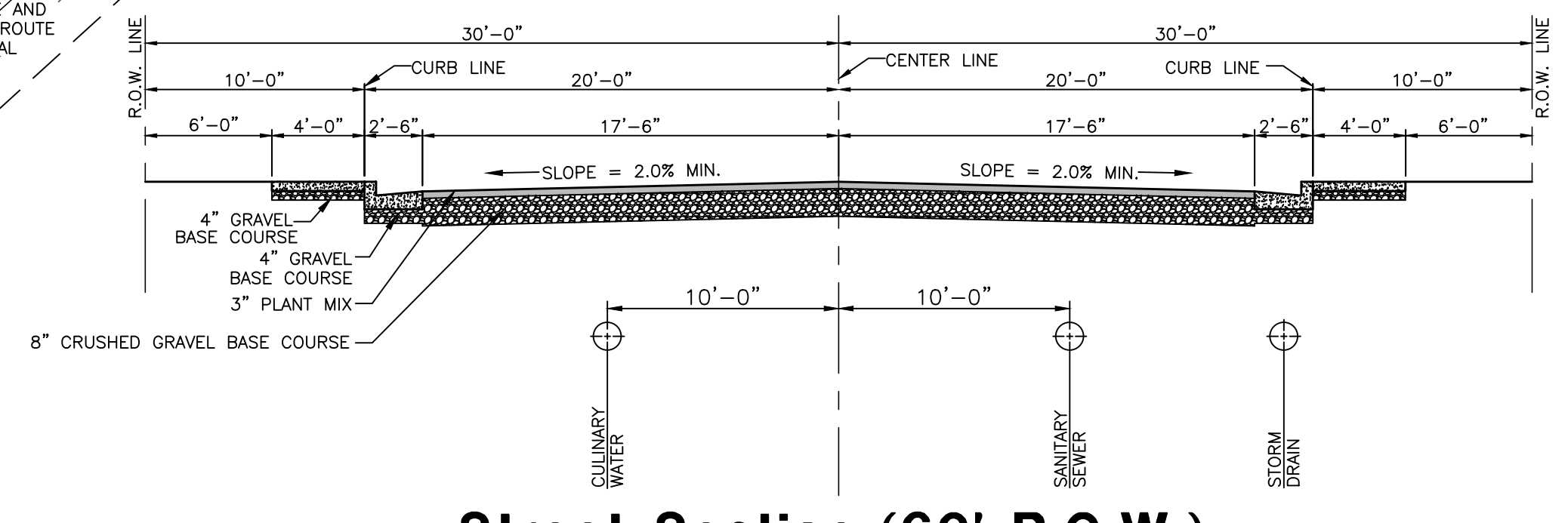
TYPICAL TOWNHOWN DETAIL
NO SCALE

LEGEND

- BOUNDARY LINE
- LOT LINE
- ADJOINING PROPERTY
- CENTERLINE
- PROPOSED CULINARY WATER LINE
- PROPOSED IRRIGATION LINE
- PROPOSED SANITARY SEWER LINE
- PROPOSED STORM DRAIN LINE
- PROPOSED MANHOLE
- EXISTING MANHOLE
- EXISTING CANAL/DITCH
- PROPOSED STREET LIGHT
- PROPOSED FIRE HYDRANT

PHASING LEGEND

- PHASE 1
- PHASE 2
- PHASE 3
- PHASE 4
- PHASE 5
- PHASE 6
- PHASE 7
- FUTURE PHASE



Street Section (60' R.O.W.)
SCALE: NONE

Delta Crossings

Delta Crossings, Millard County, Utah

Reeve & Associates, Inc.
5160 SOUTH 1500 WEST, RIVERDALE, UTAH 84405
TEL: (801) 321-3100 FAX: (801) 621-2666 www.reeve.co

LAND PLANNERS • CIVIL ENGINEERS • LAND SURVEYORS
TRAFFIC ENGINEERS • STRUCTURAL ENGINEERS • LANDSCAPE ARCHITECTS

REVISIONS	DESCRIPTION
DATE	

Delta Crossings
PART OF THE SECTIONS 5, 6, & 8, T-17S, R-6W, S.L.B. & M., U.S. SURVEY
DELTA CITY, MILLARD COUNTY, UTAH

Preliminary Utilities

Project Info.
Engineer: N. Reeve
Planner: C. Cave
Designer: E. Roche
Date: 6-21-22
Name: DELTA CROSSINGS
Number: 8059-01

Sheet	2
2	Sheets